PALANTIR

An open-source digital twin bridging astrodynamics and mission control.

APRIL 2026

Where this comes from

12 years mission-critical Java systems engineering Telco SDN · FinTech transactional systems

Currently shipping: distributed booking platform with protocol translation and a spatial data normalisation engine *(Coderama, 2022 – current)*

MSc thesis (UMB, 2017): C++/OpenMP implementation of a peer-reviewed heavy-ion event-classification method

I stopped reading about space and started building.

Why the ground segment

TELCO SDN *(where I ship)*	SPACE GROUND SEGMENT
YANG (RFC 7950)	**XTCE** (CCSDS 660.0-B-2)
NETCONF / RESTCONF (RFC 6241 / 8040)	**CCSDS Space Packet** (CCSDS 133.0-B-2)
OpenDaylight (Apache Karaf)	**Yamcs** (OSGi-style services)
Shipped to 99.999% Telco SLAs	Mission-critical TM/TC loops

Seven years of the left column made the right column look familiar.

What it is

Three pieces glued together. Fully open-source stack.

```
┌──────────────────┐                        ┌──────────────────┐
│  PALANTIR-CORE   │  CCSDS  /  UDP :10000  │ YAMCS (upstream) │
│                  │ ──────────────────────► │                  │
│  Spring Boot 3   │   18-byte Space Pkt    │   Yamcs 5.12     │
│  Java 21 / VT    │   APID 100, 1 Hz       │   UdpTmDataLink  │
│                  │                        │                  │
│  Orekit 12.2     │                        │   XTCE MDB       │
│  SGP4 / SDP4     │      UDP :10001        │   Web UI :8090   │
│  TEME→ITRF→WGS84 │ ◄──────────────────────│                  │
│                  │   Telecommand opcode   │   UdpTcDataLink  │
└──────────────────┘                        └──────────────────┘

          ▲                                          │
          │  POST /api/orbit/tle                     │  WebSocket
          │  (hot-swap TLE)                          ▼  live params
     ┌─────────┐                                ┌─────────┐
     │ OPERATOR│                                │ BROWSER │
     └─────────┘                                └─────────┘
```

Where it is today

Not a slide deck. A working system you can run today.

```
$ docker compose up --build
```

Within a minute of a warm start:

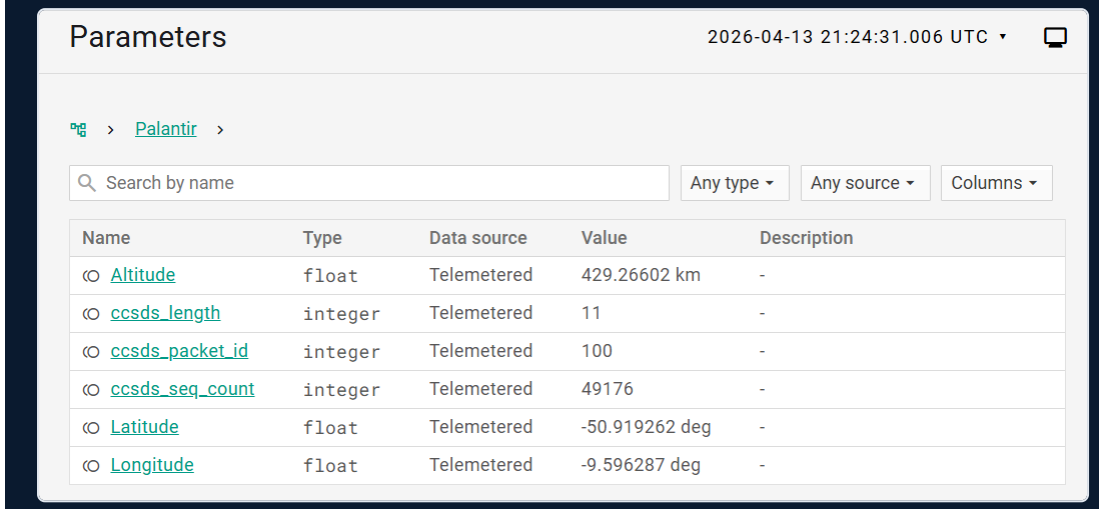
- Default ISS TLE loads automatically
- Yamcs Web UI on `http://localhost:8090`
- Three parameters streaming at 1 Hz: `/Palantir/Latitude` · `/Palantir/Longitude` · `/Palantir/Altitude`
- Persistent archive on Docker volume — survives restart

Engineering quality:

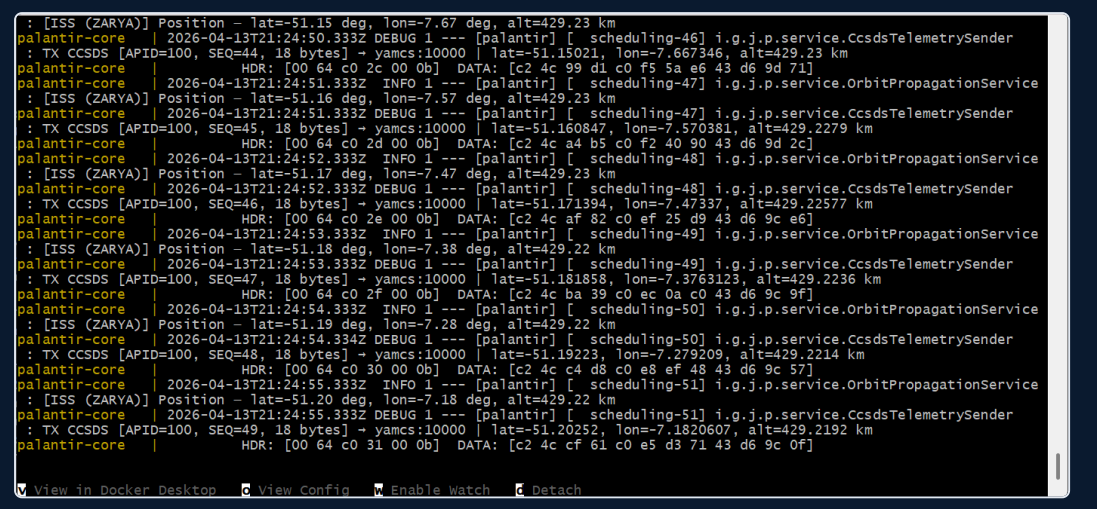
- JUnit 5 / Mockito / AssertJ — unit + Spring slice tests
- JaCoCo coverage report
- Javadoc on every main class
- 4 markdown docs in repo: README · ARCHITECTURE · FLOW · FEATURES

Active development since February 2026 · MIT license

What you'll see



Live parameters at 21:24:31 UTC



Raw CCSDS Space Packets, 1 Hz cadence

Reading the header bytes (`HDR: 00 64 c0 2c 00 0b`)

```
00 64    →  APID 100          (big-endian uint16, packet identifier)
c0 2c    →  grouping flags 11 + 14-bit sequence counter = 44
00 0b    →  data length = 11 (= payload bytes – 1, so payload is 12 B)
```

* `ccsds_seq_count` *shows the raw 16-bit field (grouping flags + 14-bit counter), so values > 16383 are expected.*

What I want to build

The Demo Day target: one end-to-end Automated Collision Avoidance flow, live on stage from a clean `$ docker compose up --build`.

1. CONJUNCTION SCREENING

- Standalone Orekit batch job · CelesTrak GP catalog (OMM XML)
- 7-day propagation window · Monte Carlo Probability of Collision
- Top events posted to the Yamcs event log

2. OPERATOR-APPROVED COMMANDING

- `FireThruster` XTCE telecommand
- Critical-significance command queue
- No autonomous burn ever flies

3. VISIBLE PHYSICS REACTION

- Operator clicks APPROVE → impulsive Δv applied to the orbit
- Ground track visibly shifts within one orbital period
- *A closed loop, made visible.*

What I do not know yet

I built Palantir to test whether twelve years of carrier-grade systems engineering can translate into the space sector. **That is precisely why I am applying.**

Two paths I would be happy with, either or both:

- **Path A — sector transition.** Use the six months as a structured entry into the European space industry. Build the network, understand how operators and integrators hire, and position myself for a role in mission-critical space software.
- **Path B — own project trajectory.** If Palantir proves viable during the programme, continue it as an independent initiative beyond September. If the traction is real, explore ESA BIC Slovakia or similar incubation as a longer-term step.

Open questions either path would help me answer:

- What are the real bottlenecks in European ground segment operations today? Where does the current toolchain break or force manual workarounds?
- Where does the open-source stack (Yamcs, Orekit, OpenC3) stop short of production needs, and what do teams build around it?
- Which institutional mission control systems are stuck on legacy tooling, and where is modernisation blocked — by budget, qualification, or risk aversion?

A stretch ambition, if the programme allows:

- Synthetic telemetry generation + LSTM autoencoder anomaly detection with ONNX export. Off the September critical path, but a track I would gladly explore.

My commitments in return:

- **Working prototype available from day one.** Live demo at every Advisory Board checkpoint, on the reviewer's laptop.
- **All artefacts open-sourced under MIT** — no IP friction, no surprises.
- **Genuinely open to mentor guidance.** The Automated Collision Avoidance flow is my current plan for Demo Day, but I will adapt direction if Advisory Board feedback points to something stronger.